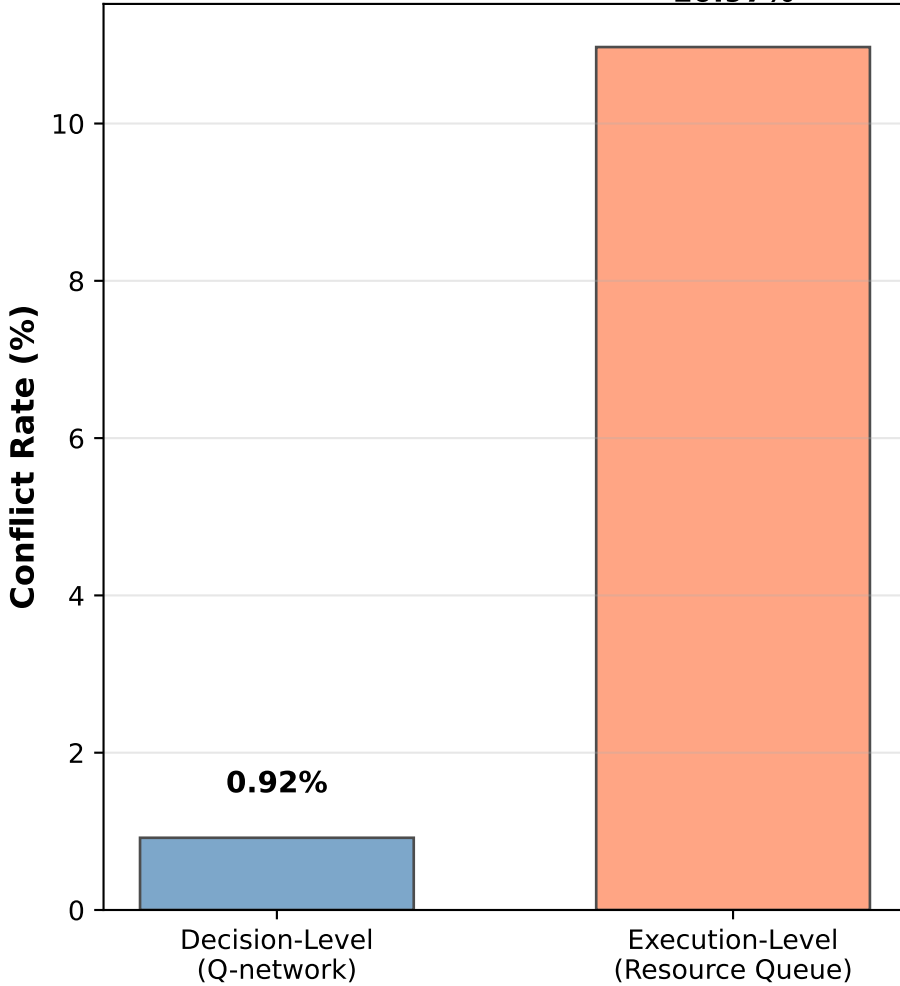
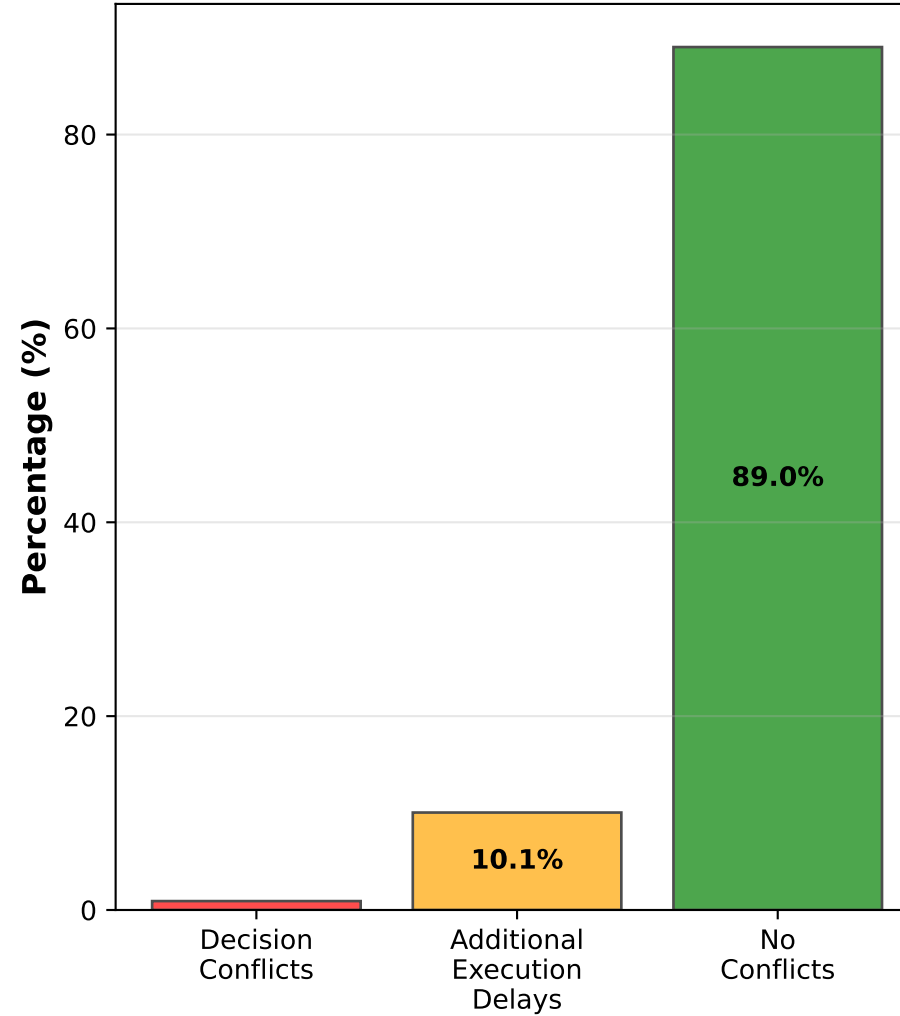


Two-Level Conflict Analysis: Decision vs Execution

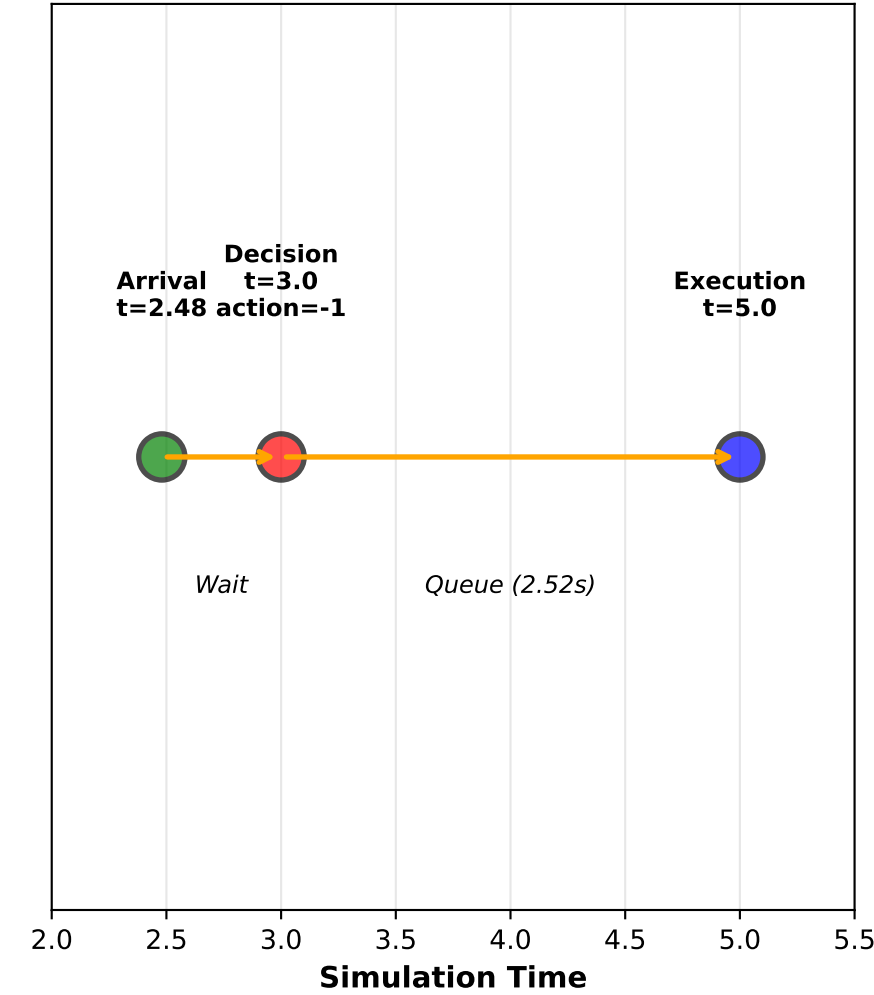
Conflict Rate Comparison



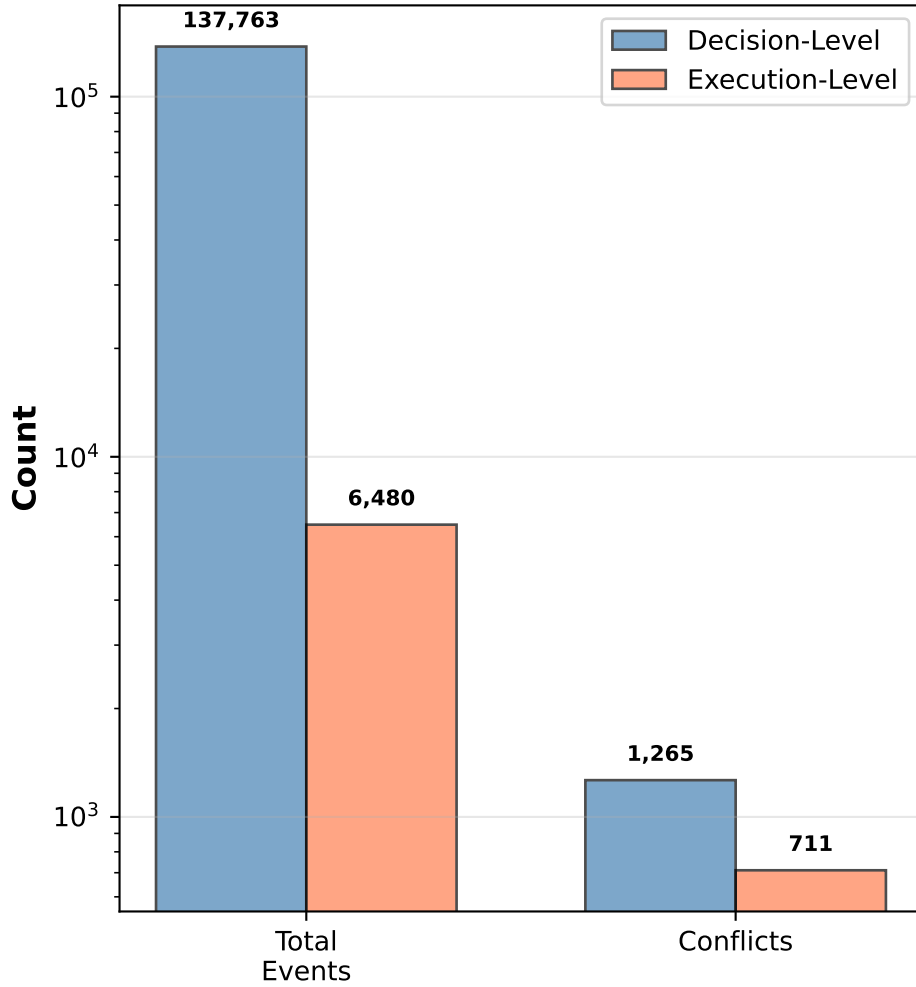
Conflict Breakdown



Episode 0, Job 4: Conflict Validation



Absolute Counts Comparison



TWO-LEVEL CONFLICT SYSTEM

Decision-Level (Agent Policy):

- Rate: 0.92%
- Count: 1,265 / 137,763
- Meaning: Agent selected unavailable action
- Mechanism: ϵ -greedy + Q-network

Execution-Level (Environment):

- Rate: 10.97%
- Count: 711 / 6,480
- Meaning: Job waited for resources
- Mechanism: SimPy queue (machine + operator)

Key Insight:

Low decision-level rate (0.92%) indicates effective Q-network learning for action selection.

Higher execution-level rate (10.97%) reflects realistic manufacturing resource contention, independent of agent learning quality.

Difference: 10.05% represents pure environment delays (operator busy, machine busy) that occur even with optimal agent policies.

Overall System Conflict Distribution

